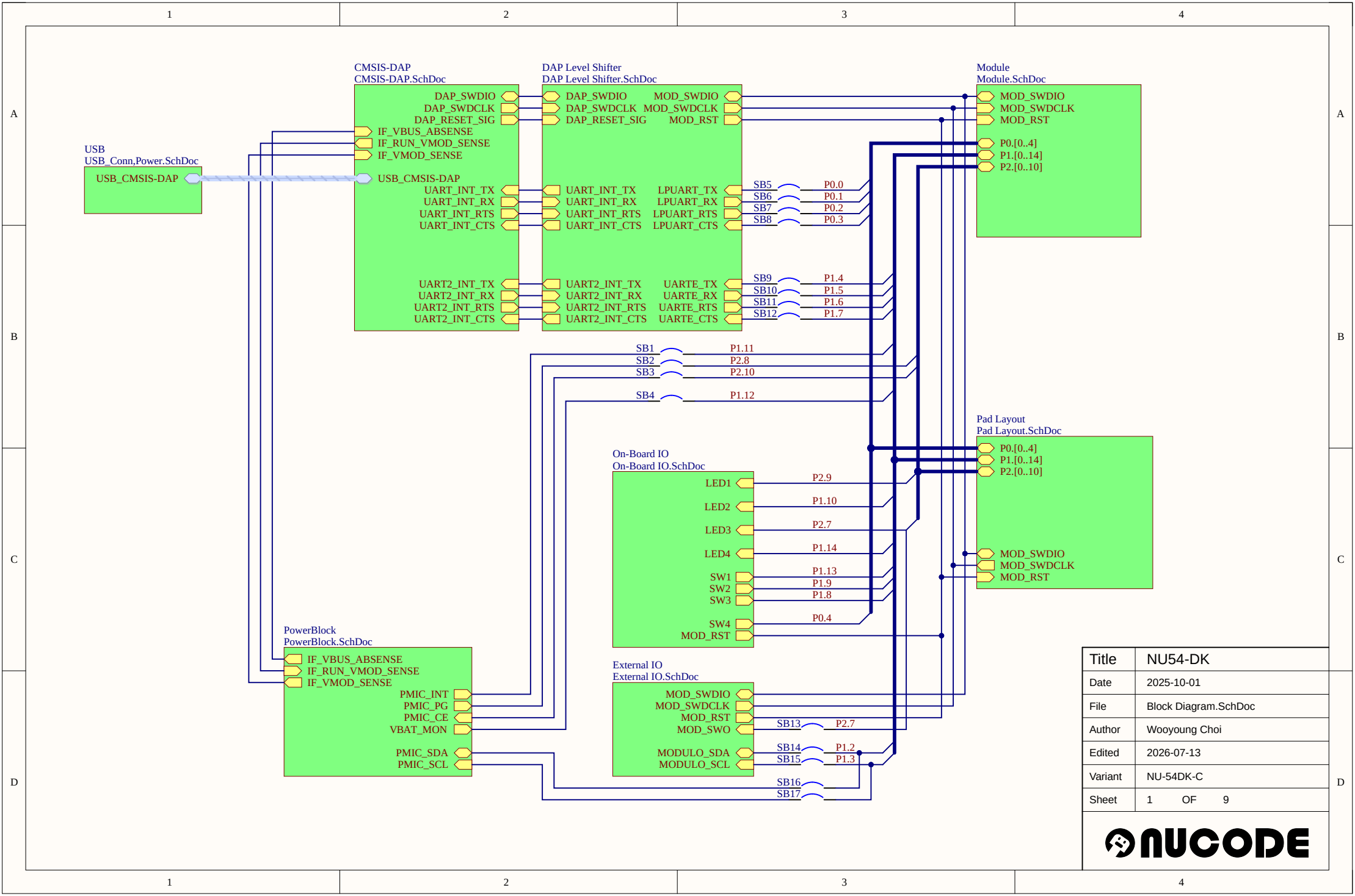


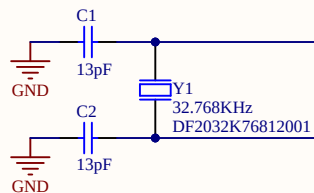


Title	NU54-DK
Date	2025-10-01
File	Block Diagram.SchDoc
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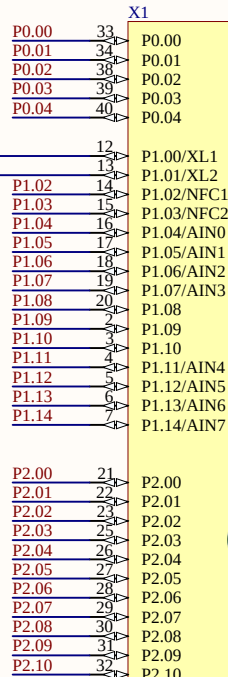


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P1.[0..14] P1.[0..14]
P2.[0..10] P2.[0..10]

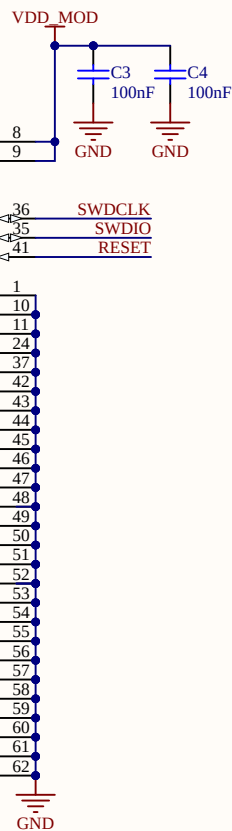
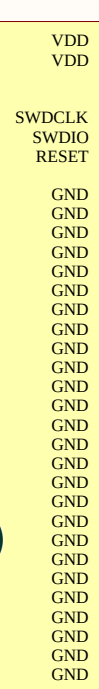
MOD_RST RESET
MOD_SWCLK SWCLK
MOD_SWDIO SWDIO



P1.00
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
P1.08
P1.09
P1.10
P1.11
P1.12
P1.13
P1.14



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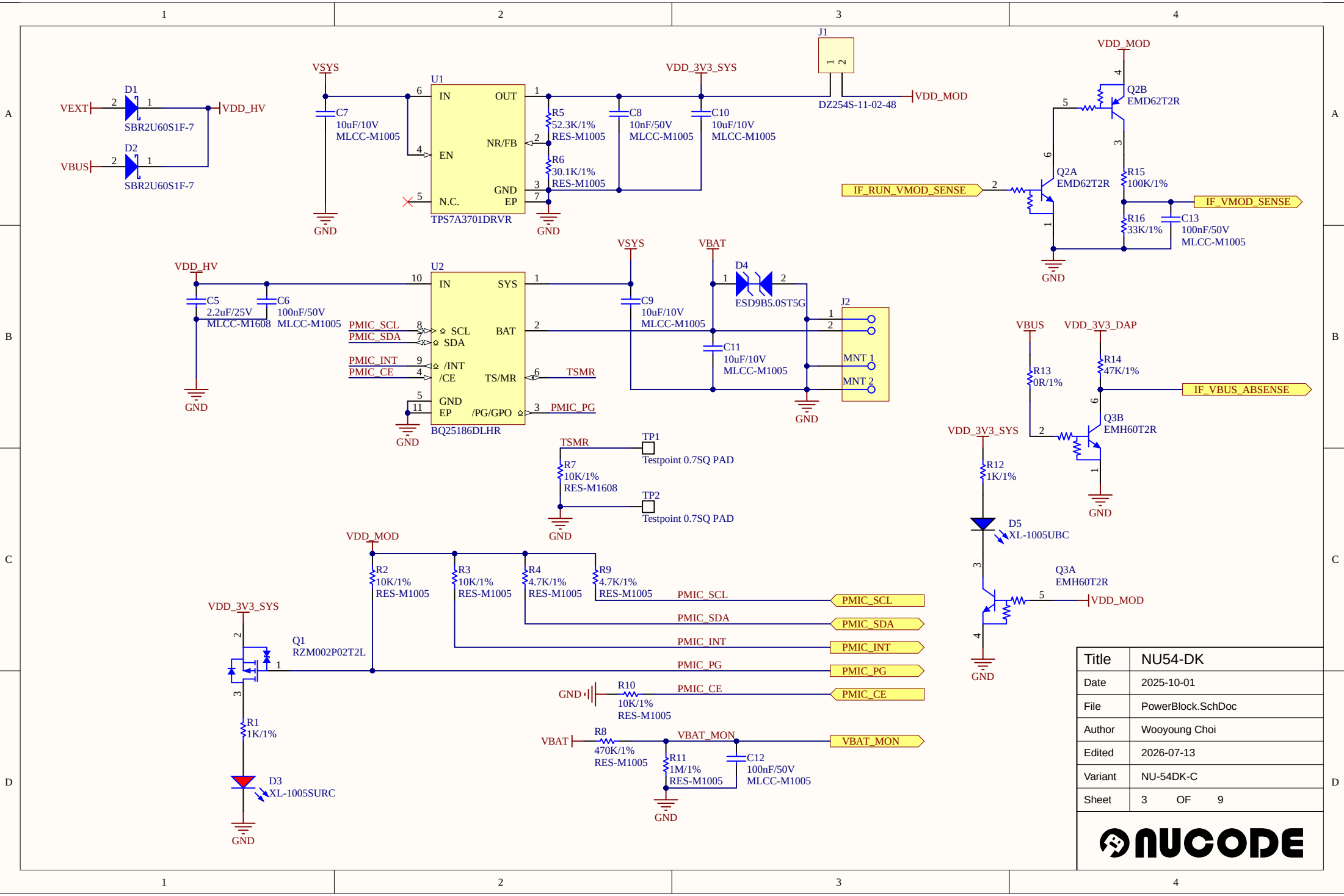


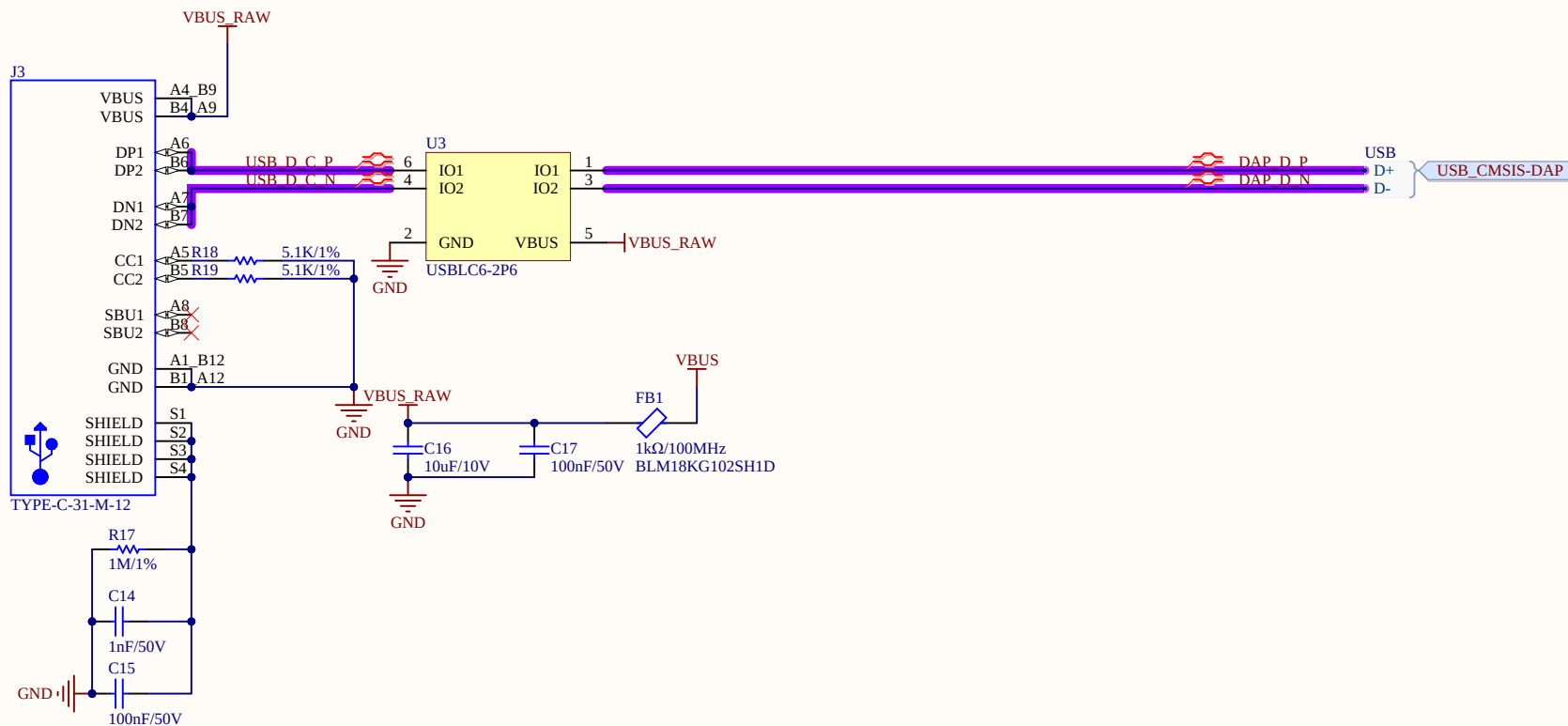
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P0.01 P0.1
P0.02 P0.2
P0.03 P0.3
P0.04 P0.4
P2.00 P2.0
P2.01 P2.1
P2.02 P2.2
P2.03 P2.3
P2.04 P2.4
P2.05 P2.5
P2.06 P2.6
P2.07 P2.7
P2.08 P2.8
P2.09 P2.9

P1.00 P1.0
P1.01 P1.1
P1.02 P1.2
P1.03 P1.3
P1.04 P1.4
P1.05 P1.5
P1.06 P1.6
P1.07 P1.7
P1.08 P1.8
P1.09 P1.9

Title	NU54-DK		
Date	2025-10-01		
File	Module.SchDoc		
Author	Wooyoung Choi		
Edited	2026-07-13		
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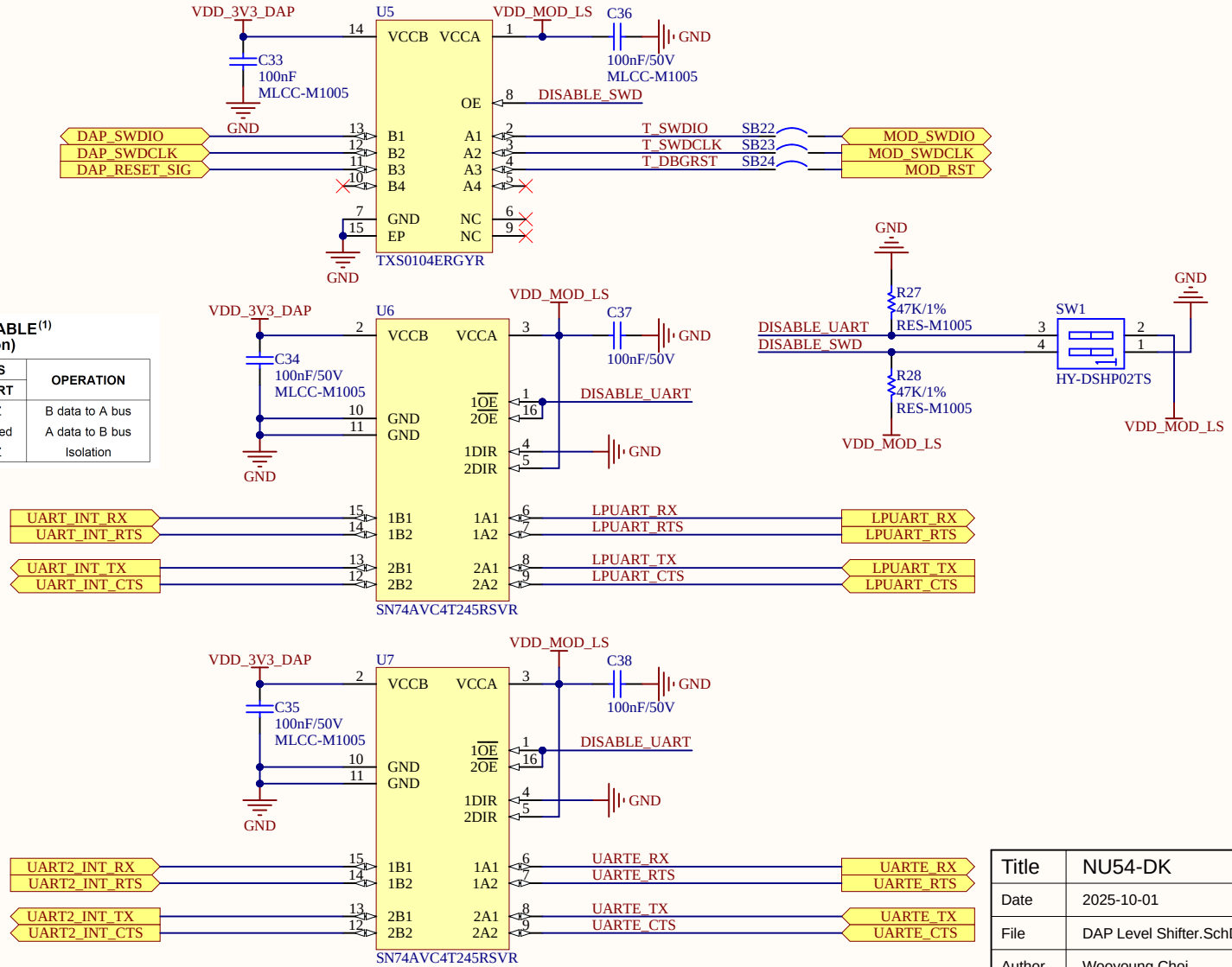
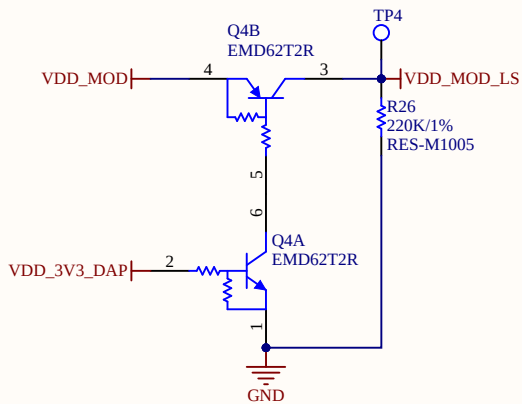




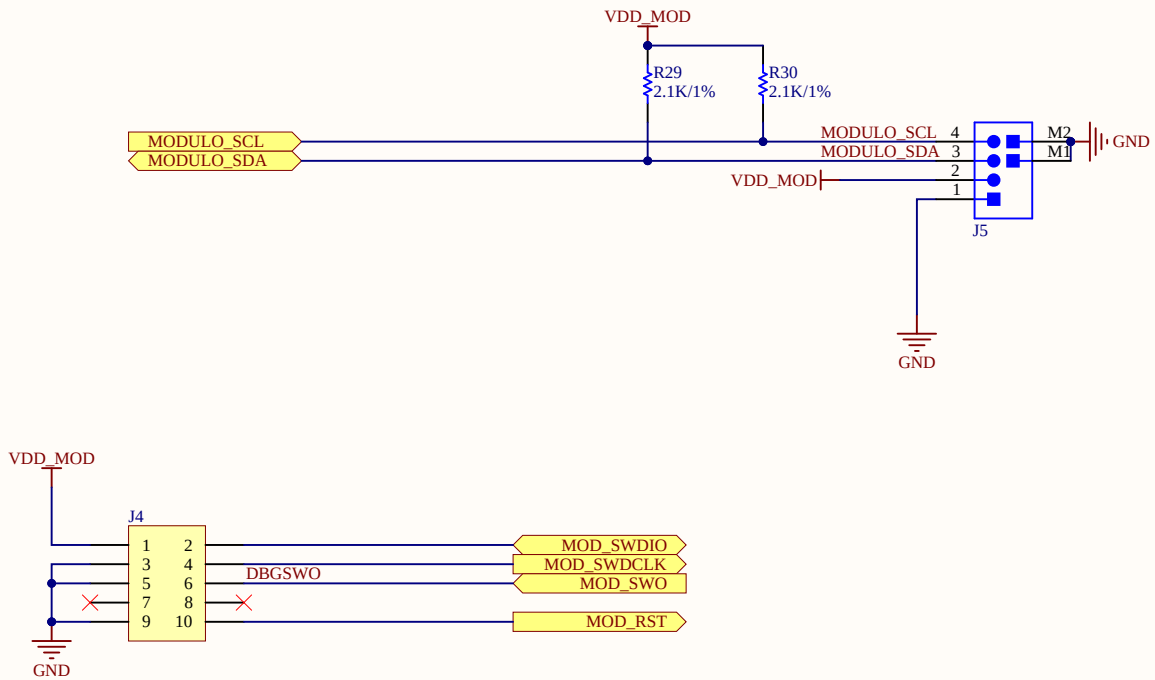
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Date	2025-10-01		
File	USB_Conn,Power.SchDoc		
Author	Wooyoung Choi		
Edited	2026-07-13		
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Table 1. FUNCTION TABLE⁽¹⁾
(each 2-bit section)

CONTROL INPUTS		OUTPUT CIRCUITS		OPERATION
OE	DIR	A PORT	B PORT	
L	L	Enabled	Hi-Z	B data to A bus
L	H	Hi-Z	Enabled	A data to B bus
H	X	Hi-Z	Hi-Z	Isolation



Title	NU54-DK		
Date	2025-10-01		
File	DAP Level Shifter.SchDoc		
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Title	NU54-DK		
Date	2025-10-01		
File	External IO.SchDoc		
Author	Wooyoung Choi		
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1

2

3

4

A

A

B

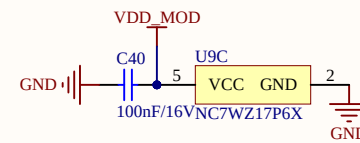
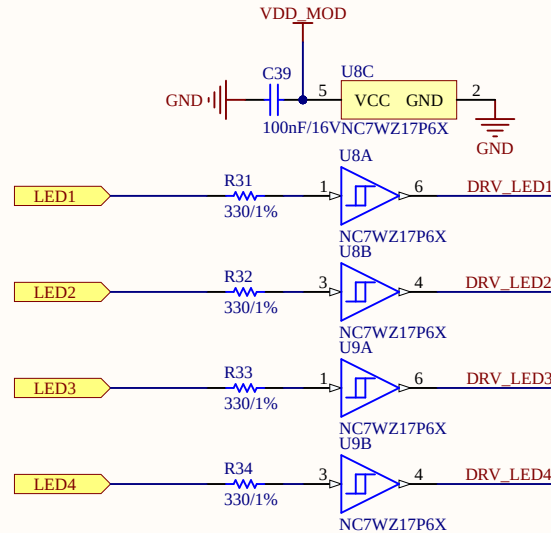
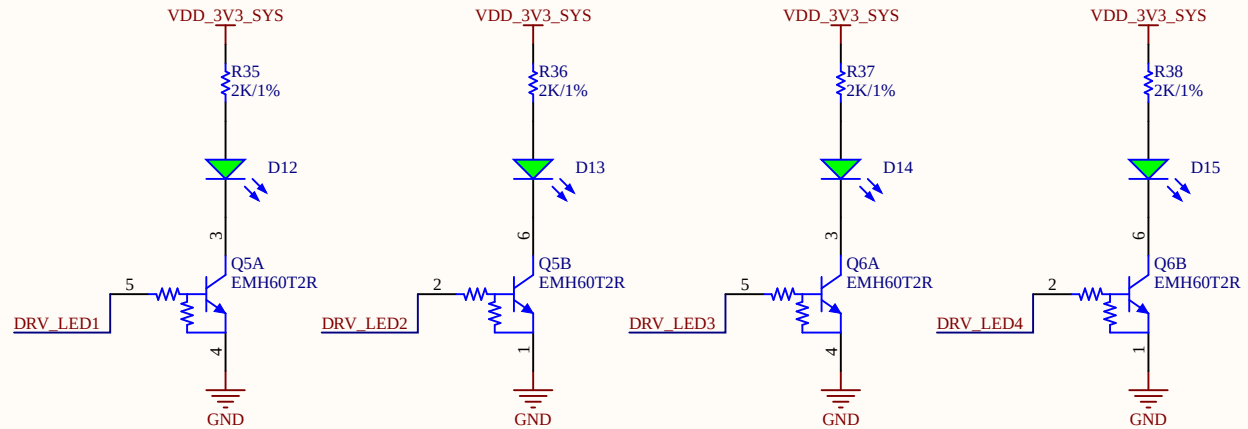
B

C

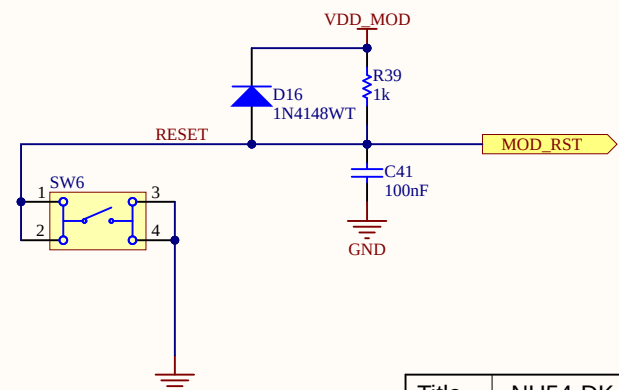
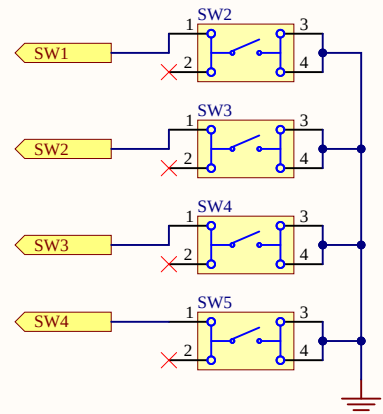
C

D

D



USE INTERNAL PULLUP



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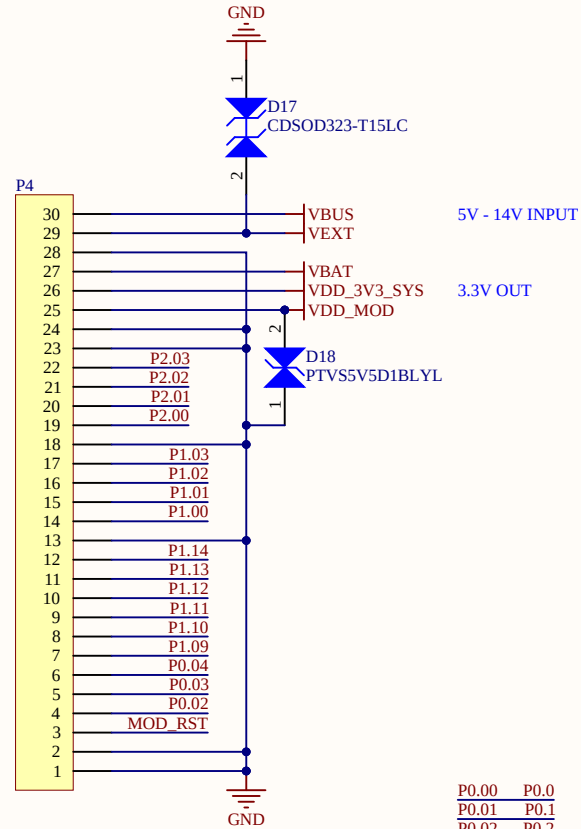
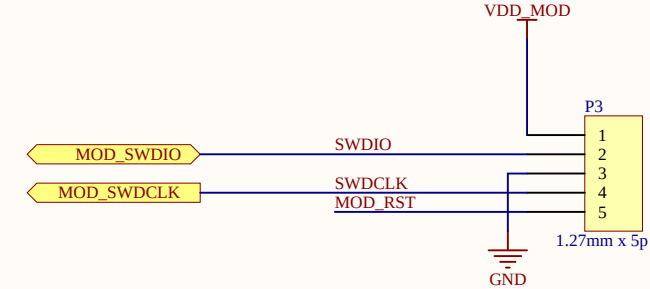
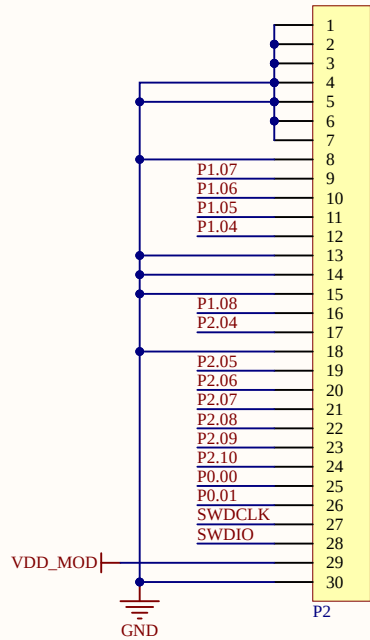
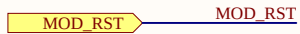
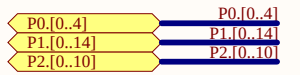


1

2

3

4



P0.00	P0.0	P1.00	P1.0
P0.01	P0.1	P1.01	P1.1
P0.02	P0.2	P1.02	P1.2
P0.03	P0.3	P1.03	P1.3
P0.04	P0.4	P1.04	P1.4
P2.00	P2.0	P1.05	P1.5
P2.01	P2.1	P1.06	P1.6
P2.02	P2.2	P1.07	P1.7
P2.03	P2.3	P1.08	P1.8
P2.04	P2.4	P1.09	P1.9
P2.05	P2.5		
P2.06	P2.6		
P2.07	P2.7		
P2.08	P2.8		
P2.09	P2.9		

Title	NU54-DK		
Date	2025-10-01		
File	Pad Layout.SchDoc		
Author	Wooyoung Choi		
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